Lingyi Zhou

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SUMMARY

AI/ML Engineer and Full-Stack Engineer with proven expertise in designing commercial-grade **Multi-Agent Systems**, **Graph RAG**, and **GenAI** solutions. Specializes in transforming complex business requirements into scalable, autonomous AI platforms. Deep experience leading the full project lifecycle—from architecting Python/Rust backends and Agentic workflows to managing CI/CD pipelines in hybrid cloud environments. Adept at bridging the gap between cutting-edge LLM research (LangGraph, MCP) and production stability.

TECHNICAL SKILLS

- **AI & GenAI:** Multi-Agent Systems, Graph RAG, AI Agentic, LangChain, LangGraph, n8n workflow, Prompt Engineering, Semantic Search, MCP (Model Context Protocol); Real-time inference; MLOps; model optimization (latency/throughput)
- **Backend & Languages:** Python, TypeScript, Rust (PyO3), FastAPI, Next.js, SQL/NoSQL.
- **Cloud & DevOps:** Kubernetes (K8s), Docker, Azure, CI/CD Automation, Microservices, Hybrid-Cloud.
- **Languages:** English (C1 - Professional), Mandarin (Native), German (A2 - Basic).

PROFESSIONAL EXPERIENCE

GREENZERO Essen, German

AI Engineer (Core Developer & System Designer) *Sep 2025 – Present*

Leading the architecture and development of the "Horus Platform," an enterprise CRM and automation system for the ecological credits market

- **Enterprise Automation & CRM:** Engineered the full-stack architecture using Next.js 16, TypeScript, and Supabase; integrated real-time data pipelines for streaming signals. Developed an enterprise-grade CRM and automation system powering AI-driven sustainability analytics for the German ecological credits market and data synchronization workflows that aggregate competitor data across multiple marketplaces to drive business intelligence.
- **n8n Workflow Automation:** Built automated workflows using n8n and LLMs, connecting HubSpot CRM, WordPress, Supabase, and SharePoint for real-time bi-directional data synchronization. Built an AI-powered procurement intelligence system that monitors markets and tenders and classifies business opportunities, reducing manual review time by 80%.
- **Enterprise RAG Knowledge Base & Workflow Automation:** Architected a private knowledge retrieval system using **RAGFlow** to process heterogeneous internal data, leveraging **Deep Document Understanding** to successfully parse complex layouts (financial tables, scanned PDFs) where traditional OCR failed. Upon this foundation, I engineered an **Agentic Workflow** for the tender process that orchestrates **Hybrid Search** (Vector + Keyword) for historical standard clauses and Web Search for dynamic client contexts. This integration transformed the drafting lifecycle from serial manual entry to a parallel "Review & Polish" model, **tripling team throughput** (from 1 to 3 tenders/cycle) while ensuring high technical accuracy through citation-based grounding.

Z. AI Beijing, China

AI Software Developer (Full-time)

Core developer for "AI City" (Multi-Agent Metaverse) and the "AutoGLM" Enterprise Agent Platform. *July 2024 – Jun 2025*

AI City Platform: Autonomous Multi-Agent Interaction Metaverse | Core Developer & System Designer

- Led the development of **AI City** using **TypeScript**, evolving pure LLMs into a visualized multi-agent simulation platform with integrated **RAG knowledge bases** and human users through **chatbot** interfaces. Deployed pilot platforms across **two cities**, each hosting **50+ enterprise agents** interacting autonomously, the platform enabled enterprises to autonomously discover business synergies and form partnerships, validating the core concept in a live simulation environment..
- Architected the next-generation **A2A multi-agent framework**, drawing from Google's A2A principles to enhance scalability, agent coordination, and cost efficiency. Designed and implemented **10+ agent types** using **MCP** and **LangGraph**, optimizing internal workflows for **host-agent** decision-making and multi-step task delegation.

"AutoGLM" Enterprise AI Agent Platform | Module Lead

- Architected and led the development of the core intelligence layer for an **enterprise AI Agent platform**, transforming user requests into automated, **multi-step workflows** executed by specialized agents..
- Built **Rust sandbox** runtime for secure code execution, replaced Docker with **OS-native isolation** (Landlock/Seatbelt + PyO3), achieving 100ms startup vs 1.5s — optimized for low-latency inference and 10+ concurrent Python/SQL interpreters per node with memory/filesystem restrictions.
- Implemented **workflow orchestrator**, tool registry for document processor (financial/tender PDF→JSON extraction), flowchart generator (text→Mermaid→SVG), data analyst (sandboxed pandas/SQL with chart generation). Migrated agent communication to **MCP** , Platform processes 1K+ tasks daily.

DataGrand Beijing, China

AI Software Developer (Module Lead)

Led the Graph RAG module for an enterprise virtual agent solution, adopted by 100+ company clients. *Jan 2023 – June 2024*

- **Graph RAG & Real-time Retrieval Development:** Led the design of a commercial **Graph RAG** system using LangChain and Microsoft GraphRAG. Combined vector indexing with knowledge graphs to improve semantic retrieval accuracy by **40%**.
- **Cloud-Native MLOps:** Built the backend infrastructure with **FastAPI** (Streaming REST APIs for real-time data and signal ingestion) and established CI/CD automation frameworks for hybrid-cloud environments (Azure & On-Premises Kubernetes).
- **Intelligent Document Extraction (IDEP):** Engineered a Zero-Shot LLM extraction service with JSON Schema enforcement and a **Human-in-the-Loop (HITL)** validation interface.
- **Impact:** Achieved **>95% accuracy** on complex document analysis and generated **¥500K+ in revenue** within the first 6 months.

EDUCATION

Yangtze University | Computer Science **China**

Bachelor June 2019 - June 2023